

LUMBAR PLEXUS

FEMORAL NERVE AND OBTURATOR NERVE

PLAN

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I. INTRODUCTION:

The innervation of the lower limb is provided by two plexuses: lumbar and sacral. These two plexuses are anastomosed by the lumbosacral trunk. The two plexuses together are referred to as the lumbosacral plexus. The lumbar plexus is an anastomotic network of nerve fibers originating from the lumbar spinal cord centers, from L1 to L4.

II. DESCRIPTIVE ANATOMY:

1. STRUCTURE:

- The lumbar plexus is formed by the union of the anterior rami of the lumbar nerves L1, L2, L3, and L4. Each of these rami anastomoses with the rami above and below it.
- The anterior branch of L1 receives an anastomosis from the 12th intercostal nerve and divides into three branches: the iliohypogastric nerve, the ilioinguinal nerve, and a branch for the genitofemoral nerve.
- The anterior branch of L2 divides into four branches to form the genitofemoral nerve, the lateral femoral cutaneous nerve, the obturator nerve, and the femoral nerve.
- The anterior branch of L3 divides into three branches to form the lateral femoral cutaneous nerve, the obturator nerve, and the femoral nerve.
- The anterior branch of L4 divides into three branches to form the femoral nerve, the obturator nerve, and the lumbosacral trunk.

2. SHAPE AND LOCATION: The lumbar plexus is triangular. It is located between the bundles of the psoas major muscle, running along the lateral faces of the vertebral bodies, in front of the costiform processes.

III. COLLATERAL BRANCHES:

It gives off muscular branches to the quadratus lumborum, psoas minor, and psoas major muscles, as well as nerves: the iliohypogastric nerve, the ilioinguinal nerve, the genitofemoral nerve, and the lateral femoral cutaneous nerve.

1. Iliohypogastric Nerve:

This is a mixed nerve originating from the anterior branch of the first lumbar spinal nerve (L1). It runs downward and laterally, crossing the anterior surface of the quadratus lumborum muscle, then passes between the transversus abdominis and internal oblique muscles.

At the level of the anterior superior iliac spine, it divides into two terminal branches: the abdominal and the genital branches.

It innervates: the skin of the lower abdominal wall, the genital region and the superomedial part of the thigh, the muscles of the lower abdominal wall.

2. **Ilioinguinal nerve:** A sensory nerve located below the iliohypogastric nerve; its origin, course, termination, and terminal branches are identical to the iliohypogastric nerve.

3. Genitofemoral nerve:

This mixed nerve arises from the anterior branches of the first and second lumbar spinal nerves (L1 and L2). It passes through the psoas muscle, descends along its anterior surface, and terminates at a variable distance from the inguinal ligament, dividing into two terminal branches: the lateral femoral and the medial genital. It innervates the genital region, the skin of the femoral triangle, and the abdominal muscles.

4. Lateral cutaneous nerve of the thigh:

This exclusively sensory nerve arises from the anterior branches of the second and third lumbar spinal nerves (L2 and L3).

It runs downwards and outwards between the two bundles of the psoas major muscle, continuing its course towards the anterior superior iliac spine and the inguinal ligament. It then travels beneath the inguinal ligament, crosses the sartorius muscle, and in the anteromedial region of the thigh, it pierces the fascia lata and becomes superficial. It divides into two terminal branches:

- A gluteal branch: supplying the skin of the lateral region of the buttock
- A femoral branch: supplying the skin of the anterolateral region of the thigh.

IV. TERMINAL BRANCHES :

1) Obturator nerve:

- Origin: This mixed nerve arises from three roots originating from the anterior branches of the 2nd, 3rd, and 4th lumbar spinal nerves (L2, L3, L4).
- Course: It crosses the sacroiliac joint, passes into the pelvic cavity, and enters the obturator foramen where it terminates by dividing into two terminal branches.
- Terminal branches:
 - Anterior branch: This superficial branch enters the medial compartment of the thigh and gives off three branches to the gracilis, adductor longus, and adductor brevis muscles, as well as a cutaneous branch.
 - Posterior branch: deep, it also penetrates the medial muscular compartment, then lies behind the adductor brevis and in front of the adductor magnus. It descends to the adductor hiatus, which it crosses to reach the popliteal fossa.

It gives off muscular branches to the superior and middle fibers of the adductor magnus and to the obturator externus, and an articular branch to the knee joint.

- Function
 - Motor: adduction and lateral rotation of the thigh; it innervates the following muscles: obturator externus, pectineus, adductor brevis, adductor longus, adductor magnus, and gracilis.
 - Sensory: it provides sensation to the skin on the medial aspect of the thigh.

2) **Femoral Nerve:**

- **Origin:** Mixed nerve, arising from three roots originating from the anterior rami of the 2nd, 3rd, and 4th lumbar spinal nerves (L2, L3, and L4) between the two planes of the psoas major muscle.
- **Course and Relationship:** The femoral nerve lies anterior to the iliopsoas muscle, passes under the inguinal ligament, and terminates in the femoral triangle, dividing into four terminal branches.
- **Terminal Branches:**
 - Lateral musculocutaneous nerve: Mixed nerve, it divides into muscular and cutaneous branches.
 - Muscular branches: Supply the sartorius muscle.
 - Cutaneous branches: Innervate the skin of the anterior thigh.
 - Medial musculocutaneous nerve: It divides into muscular and cutaneous branches.
 - Muscular branches: supply the pectineus and adductor longus muscles.
 - Cutaneous branches: innervate the skin on the anteromedial aspect of the thigh.
 - Nerve to the quadriceps femoris: it is exclusively motor and gives off four branches for the four heads of the quadriceps femoris muscle: nerve to the vastus intermedius, nerve to the vastus medialis, nerve to the vastus lateralis, and nerve to the rectus femoris.
 - Saphenous nerve: it is exclusively sensory, descends along the femoral artery, passes through the adductor canal, and reaches the joint line of the knee where it divides into two terminal branches: one patellar and the other tibial.
- **Innervation territory:**
 - Motor: the femoral nerve flexes the thigh and extends the leg; it provides motor function to the following muscles: iliopsoas, pectineus, adductor longus, quadriceps femoris, and sartorius.
 - Sensory: it provides sensitivity to the skin in the following areas: The anterior thigh, the anteromedial surfaces of the knee, leg and neck of the foot.

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